

A zebrafish eyeball, moved by six muscles

Progress note — written to be read, not decoded

3 August 2026

prototype/eye/. Operational summary: STATUS.md. Plain English throughout.
Organised by phase, because phases are where we stop and you decide. Each phase is two boxes: what we *set out* to do, and what came back when we *ran* it.

1. What this is

A deformable zebrafish eyeball in a bony orbit, rotated by six extraocular muscles that are themselves contracting MLS-MPM bodies, expressed entirely as a Plexus2 `spec.yaml`.

Nothing applies a force “to the eye”. A muscle shortens because an active stress acts along its fibres; it cannot reel itself up because its origin end is anchored to bone; and it moves the globe because its tendon end is embedded in the sclera and the two bodies share one background grid. **The globe rotates because a muscle got shorter.** That is the whole mechanism.

Nothing was promoted. Eleven operator contracts and one new *implementation* of an existing one live in `eye_ops.py` and `muscle_ops.py`; importing them registers them, and the run is executed by the stock `plexus.schema.load` and `plexus.engine.run`. `src/plexus` is untouched. The deliverable is a `spec.yaml`.

The decomposition into sets

set	what it is	state
eye	the globe, as one organ	<code>pos</code> (centroid), <code>gaze</code> = (horizontal, vertical, torsion), degrees
<code>mpm_particle</code>	the globe’s material points (parent <code>eye</code>)	<code>pos/vel</code> + F , C , mass, Lamé, volume
<code>muscle</code>	the six extraocular muscles	<code>act</code> (the only integrated block), <code>tension</code> , <code>length</code>
<code>muscle_particle</code>	each muscle’s material points (parent <code>muscle</code>)	as <code>mpm_particle</code>
<code>orbit</code>	the bony socket	<code>pos</code> , the centre of the cup
<code>mpm_grid</code> (field)	one background grid, shared by both bodies	—

The shared grid is the design. It is what mechanically couples a contracting muscle to the sclera, and it is free — sharing a background grid is simply how MLS-MPM couples any two bodies. The stock `mpm_scatter` *overwrites* the grid, so a second *implementation* of that same contract, `implementation: accumulate`, was registered: identical biology, different numerics. Per Plexus2 that is the correct way to extend — never an edit to the engine.

The operators

operator	kind	what it does
eye_anatomy	rewire	squash the seeded ball into the ovoid; label sclera / cornea / iris / pupil / choroid / vitreous / lens; per-region Lamé
muscle_morphogenesis	rewire	shape each muscle's points into a tapered strap along its arc of contact; fibres; bone cap and tendon cap
eye_pose	aggregate	Kabsch fit of the shell against its rest state → gaze and torsion
muscle_geometry	aggregate	bin points by fibre coordinate → centreline, length , insertion, line of action, rotation axis
oculomotor_drive	lateral	gaze program → PID → rectified projection onto each muscle's axis → activation dynamics
muscle_contract	exchange	$\sigma_a = A a (1 + \beta(\lambda - 1)) \mathbf{f}\mathbf{f}^\top$ along the fibre
bone_anchor	lateral	pin the origin cap to the skull
muscle_sleeve	lateral	the orbital pulley: a purely <i>transverse</i> restoring force, so the path is held but shortening is free
orbit_socket	lateral	penalty contact with the spherical cup + orbital-fat suspension
mpm_scatter [accumulate]	exchange	the second body into the shared grid

The schedule closes a loop: pose readout → gaze error → innervation → contraction → MLS-MPM → motion → pose readout.

2. Where we actually are

phase	state	what it produced
0 — build the plant	done	25 trials, 4 named ceilings, a working point, a metric suite
1a — differentiable identification	FAILED, AND KEPT	a surrogate that fails its own fidelity test, and refuses to report
1b — the dynamics-watching critic	FAILED, AND KEPT	a VLM that reads a rotating eye as static
2 — PID autotune by step injection	in flight	the experiment that decides control <i>vs</i> mechanics

Two automatic instruments were built to judge the plant and **both failed**. That is why Phase 2 exists, and why it is an *open-loop measurement* rather than another optimisation.

The archive was pruned to the Phase-2 baseline. Only t01, t02 and t03_c_a survive on disk; the run folders from t04 onward were deleted after the campaign closed. Their metrics are in the git history and are quoted here as the record.

3. Phase 0 — Build the plant

Phase 0 — define

done

Goal. A globe that deforms a little but does not come apart, six muscles that visibly contract, a skull that keeps the eye in its socket, and a movie showing deformation, stress and activation — all of it a `spec.yaml`, with the operators registered from the prototype folder and the engine untouched.

The constraint that changed the model. The first version had a muscle as a *line of action*, its pull a body force injected at the insertion patch. Nothing contracted, nothing deformed, and the tendon’s grip on the sclera was *asserted*. Making the muscles tissue — their own particle set, coupled through the shared grid — is what turned a picture into a mechanism.

What “done” means, written as a test. `run_eye.verdict`: finite; reaches its commands within 6°; torsion > 6°; horizontal range > 35°; recruitment 4/4; centroid drift < 6% of the globe radius; globe strain₉₉ in [0.004, 0.12]; peak shortening in [4%, 38%]; globe radius worst < 6% and roundness spread < 9%.

Phase 0 — run

25 trials, 4 ceilings

Each ceiling was a real mechanism, not a tuning miss. They are worth more than the working point.

ceiling	what it turned out to be
A and E are not independent (t02)	A muscle shortens until passive tension balances active stress, so steady shortening is set by A/E <i>alone</i> while delivered force is $A \times \text{cross-section}$. Raising A by itself: 88% shortening, globe crushed .
A/E is a cliff, not a trade-off (t03–t04)	0.25 clean, 0.30 collapses — a <i>buckling</i> threshold. But the stiff passive element that bought stability made the antagonist so stiff it ate most of the agonist’s force: the eye capped at 11° .
Length buys gaze (t06–t08)	$\theta_{\max} \sim (A/E)(L/R)$. Truncating each muscle to 55% of its path — a convenience taken so the four recti would not project onto one blob — had left the recti at $L/R = 2.3$ and the obliques at 1.15, against ≈ 3.3 in a real orbit. Restoring the length and spreading the origins around the annulus of Zinn (a <i>ring</i> , not a point): $12.7^\circ \rightarrow 29.6^\circ$.
The missing mechanism was the pulley (t15–t24)	A free-floating strap is a column in compression the moment it contracts. <code>muscle_sleeve</code> restores only the component <i>perpendicular</i> to the local fibre, tapering to zero over the distal third — Demer’s active pulley, which is exactly a transverse constraint. Everything else held fixed, it raises the buckling ceiling from ≈ 0.25 to between 0.28 (clean) and 0.32 (folds) .
and a material error (t18)	The vitreous at $E = 9$ is a shear modulus of 3.8 — very nearly a fluid. The fixed-corotated law has no failure criterion, so under a hard tendon pull the interior flowed and the globe came apart . Interior raised to 45 / 130 against a 420 sclera: same $\approx 9 \times$ core-to-shell contrast, enough shear modulus <i>everywhere</i> to hold together.

Working point (t23/t25, the full 1830-frame programme): range 33.5/18.3/7.5°, shortening 24–31% with no buckling, centroid drift 1.8% of the globe radius, strain₉₉ 0.112, and the globe’s own radius mean -0.04% , worst 0.22%, roundness spread 0.86%. *The eye deforms locally and holds its shape globally* — which is the whole reason to use MPM here, and is now a number rather than a look at the movie.

Six of nine acceptance checks pass. `reaches_commands` fails (max settled error 17.1°); `wide_gaze_range` fails narrowly (33.5° against a 35° bar); and `correct_recruitment` fails *on the atlas programme only*, at 14/26 — that programme’s 26 holds include diagonal combinations for which no single muscle is “the” expected agonist, so a top-1 test is the wrong test

there. On the four-hold probe programme it is 4/4 in every mechanically-stable trial. The real gap is the first one: the eye settles at roughly 60% of a 25° command, and everything from here is about that.

A lesson about pictures. The muscle truncation was made so the anatomy would *render* well. It cost 17° of gaze. A convenience taken for the figure was a mechanical error, and nothing in the pipeline would have caught it — the movie looked better after the change.

4. Phase 1 — Can an automatic instrument judge this?

Two were built. Both failed, and both failures are kept because they are the reason Phase 2 is shaped the way it is.

Phase 1a — differentiable identification and gain tuning

Phase 1a — define

done

Claim to test. The gap between commanded and achieved gaze is a *controller* problem, and gradient descent on the gains will close it.

Method, and why not the obvious one. The engine can keep the tape (`run(..., grad=True)`), but a rollout of 45k+13k material points $\times$ 25 substeps $\times$ hundreds of frames unrolls into a graph whose memory scales as points $\times$ steps — exactly the cost Plexus 2 says makes a global loss impractical. `../inverse_slime/` gets around it by refusing to roll out at all, so `tune_gaze.py` does the same in two stages: identify $\ddot{\theta} = Bu - C\dot{\theta} - K\theta$ per axis by ridge least squares from an archived run, then run Adam on $(k_p, k_i, k_d, \text{gain}, \text{tonic})$ over that differentiable surrogate, and verify back in the full simulation.

Success looks like a surrogate that reproduces the run it was fitted to, and a tuned gain set that measurably improves tracking in the real simulation.

Phase 1a — run

the instrument failed its own check

First the derivatives, not the model. R^2 on $\ddot{\theta}$ came out 0.28–0.40. Differentiating a strided, noisy signal *twice* amplifies its noise by $\sim h^{-2}$; reading the derivatives off a local Savitzky–Golay polynomial instead raised it to **0.48–0.72** with no change to the model at all.

Then the model. Tuning still bought 1.008 $\times$. The first explanation offered — “the drive must be saturated” — was *wrong*, and the tool’s own diagnostic said so: the agonist sits above 0.98 on only 6% of frames. So the surrogate was asked to reproduce the excursion the real eye actually made, under the gains it actually used:

peak excursion	horizontal	vertical	torsion
real	17.0°	9.0°	2.8°
surrogate	1.7°	3.0°	2.2°

The instrument failed its own check, so the measurement was thrown away. A good R^2 on $\ddot{\theta}$ certifies the *transients*; the standing error we wanted to tune away lives in the steady state $\theta_{ss} = Bu/K$, and the fit misses the real excursion by a factor of ten in the horizontal. **tune_gaze** now runs this fidelity test and **refuses to report a tuned number when it fails** — the discipline Plexus2 puts first: verify the instrument before trusting the measurement.

Two defects in the method itself, admitted here because Phase 2 exists to fix them.

1. **Closed-loop identification bias.** The fit was taken from a run *with the controller closed*. In that data the input u is a function of the state — that is what feedback means — so the regressor is correlated with the residual and least squares is biased. This is a textbook error and is very likely why B/K predicted 1.7° where the eye did 17° .
2. **The saturation statistic was the wrong statistic.** “6% of frames” was measured over *all* frames, most of which are transitions or primary-position holds where the error is near zero. The number that matters is saturation *of the agonist, during the settled tail of a hold*; with $k_p e = 0.11 \times 10^\circ$, times gain 1.9, clipped at 1.0, it is very likely pinned there. **So the conclusion “the residual is mechanical, not control” was not established.** It rested on a broken surrogate and a badly-chosen statistic, and this note previously stated it with more confidence than the evidence carried.

Phase 1b — the dynamics-watching critic

Phase 1b — define

done

Plexus2 names a *dynamics-watching critic* as a required agent, on the grounds that a worker agent reads a static strip but cannot watch a movie. That is cheap to test here, so: show the local Gemma-4-12B captioner the anterior panel, cropped so that no axis label, legend or title can leak the answer, and ask what happens. **Success is the word “rotating”.**

Phase 1b — run

it reads a rotating eye as static

Whole 1830-frame run. It read the anatomy well — “a large, light-grey circular ring”, “four small yellow clusters arranged symmetrically around the inner black core”, “elongated appendages in blue, red, purple and green”, even the dashed cup outline — and concluded: “*The overall configuration remains static throughout the sequence of keyframes.*” It never used the word eye.

That could have been a sampling artefact, since the programme returns to primary position between oscillation blocks and evenly-spaced keyframes may all land on one pose. So:

One excursion only — a 1.8s clip verified from the recorded trace to span $0 \rightarrow +14.8 \rightarrow -16.3^\circ$. This time it counted *six* appendages correctly, and it did report a change — the wrong one: “*the color of the appendage on the right side shifts from a dark olive green to a bright yellow ... the other colored appendages remain relatively stable in their positions ... the central black void remains constant.*”

The paper’s claim, isolated. The colour shift it caught is the medial rectus brightening with activation — a cue encoded *within* each frame. The 31° rotation, encoded *across* frames, it missed entirely, calling the pupil stationary while the pupil swept a third of the globe. The failure is not competence at seeing; it is competence at seeing *change*. A keyframe caption pipeline is structurally unable to referee a rotation, and no prompt fixes it. Captions in `archive/vlm/`.

Small irony worth keeping: the gold iridophore flecks exist *because* a round pupil on a rotating sphere has almost no per-frame signature. The VLM saw the flecks and never tracked their angle.

5. Phase 2 — PID autotune by open-loop step injection

Phase 2 — define

in flight

The question, stated so it can come out either way. Is the standing gaze error a *control* problem or a *mechanical* one? Phase 1a asserted an answer with a broken instrument. This phase measures it.

Method — the standard autotune, which is what Phase 1a should have been.

Open the loop and inject a step on *one muscle at a time*:

- a new operator `muscle_probe` writes activation from a prescribed waveform and ignores the gaze error entirely, so the input is no longer a function of the state and the closed-loop bias of Phase 1a is gone by construction;
- six runs, one per muscle: settle at tonic, step to a_{hi} , hold, release;
- each yields a 3-vector gaze response, so the six together give the full 3×6 **static gain matrix** $G_{km} = \Delta\theta_{k,\infty}/\Delta a_m$ read straight off the plateau — the quantity the acceleration fit had no way to see — plus the rise time and overshoot per channel.

Baseline: `archive/t03_c_a`, loaded from its own `spec.yaml` rather than rebuilt, so the probe measures exactly the archived configuration. It is the right baseline because it is *stable*: 4/4 recruitment, strain₉₉ 0.052, 17–27% shortening, no buckling, no sleeve, $A = 60$, $E = 240$, and a modest 12.7° range. A clean step response wants a well-behaved plant, not a strong one.

The prediction, written down before the run. Extrapolated to full activation, the lateral rectus static gain $\Delta\theta_{h,\infty}$ on `t03_c_a` will land in 8° – 16° — *below* the 26° it is being commanded to reach. If that holds, this configuration is mechanically incapable of the command and no PID retune can close the gap.

The falsifier. If $\Delta\theta_{h,\infty} \geq 26^\circ$, the plant has the authority and the loop is throwing it away — the gap is control, Phase 1a’s conclusion was wrong, and the autotune should recover most of it.

Secondary prediction. LR and MR will show a nearly pure direct-axis gain (off-axis $< 15\%$), while SR, IR, SO and IO will show substantial cross-axis terms — the same primary/secondary/tertiary structure the geometry already predicts, now measured in degrees-per-activation instead of as an axis direction.

Then, and only then, tune. The identified model is a handful of numbers, so the gains come out in milliseconds — no gradient through MPM needed. Two things to get right when

it happens: the actuator saturates and the drive is rectified, so this is a *Hammerstein* system (static nonlinearity then linear dynamics) and the nonlinearity must be fitted explicitly rather than absorbed into the linear part; and the integrator needs anti-windup, which `i_clip` provides but which has never been exercised.

Where differentiability actually earns its keep. Not here. The reduced plant is fitted, not differentiated. The tape is worth its cost for tuning the *anatomy* — insertion angles, arc of contact, pulley placement — where there is no reduced model to fit. That is the more interesting inverse problem and it is untouched.

Phase 2 — run

NOT STARTED

Six step-response runs on `t03_c_a`, the static gain matrix, and the verdict on control-*vs*-mechanics. Not yet run; this box is where the answer goes, including if it contradicts the prediction above.

6. The one result worth defending

No muscle action is tabulated anywhere in this prototype. Each muscle’s rotation axis is re-measured every frame from where its tissue actually is, as $\hat{n} \times \hat{u}$ (insertion radius $\times$ line of action), and `oculomotor_drive` simply rectifies the projection of the desired rotation onto it — Sherrington’s law of reciprocal innervation, one line of code. So the textbook primary/secondary/tertiary actions have to *emerge* from where each muscle inserts and where it pulls from. They do:

muscle	measured axis (x, y, z)	reading	textbook
LR	(0.03, 1.00, -0.05)	pure abduction	abduction only
MR	(0.02, -1.00, 0.03)	pure adduction	adduction only
SR	(-0.90, -0.17, 0.39)	elevation > intorsion > adduction	same order
IR	(0.90, -0.22, -0.37)	depression > extorsion > adduction	same order
SO	(0.61, 0.14, 0.78)	intorsion > depression > abduction	same order
IO	(-0.66, 0.18, -0.73)	extorsion > elevation > abduction	same order

All six agree, in sign and in rank. The medial tilt of the orbital apex is what gives the vertical recti their torsional and horizontal components; a co-axial apex makes SR and IR pure elevators and depressors, which is anatomically wrong. Recruitment measured from the runs agrees with this table on 4 of 4 commands in every mechanically-stable trial of the probe programme. *Not* on the atlas programme (14/26) — but that is a defect of the test, not of the plant: half its holds are diagonal commands recruiting two muscles at once, against which a top-1 comparison is meaningless.

Why this is worth more than a nice movie. It is a falsifiable prediction that geometry alone determines the actions, tested against a table nobody consulted while writing the code. It is also the Plexus point in miniature: the mechanism is in the operators and the anatomy, not in a lookup table inside a simulator. Phase 2 sharpens it — the same claim, in degrees per unit activation.

7. Two modelling decisions that are not obvious

The globe is an ovoid; the cup is a sphere. Teleost eyes are flattened along the optic axis. The `mpm_particle` entity can only seed a uniform ball, so the rest configuration is that ball affinely squashed, $z \rightarrow kz$ with $k = 0.82$. An affine image of a uniform ball is a uniform ellipsoid, so density stays uniform, and because this is an *initial condition* and not a deformation, F stays the identity — the ovoid is undeformed and carries no residual stress. The bony cup stays spherical with a radius just above the equatorial semi-axis: an ovoid cannot rotate inside a socket that matches it, and in the animal the gap is filled by orbital fat, not bone.

The fat exerts no torque. The retrobulbar fat pad is a restoring force toward the socket centre applied *uniformly* to every point of the globe. A uniform body force produces no torque about the centroid, so the fat recentres the eye without ever resisting gaze — exactly the division of labour the anatomy has. Without it, the residual normal component of muscle tension slowly walks the globe around inside the cup.

Both ends of every muscle are fixed, by two different mechanisms. The origin cap is pinned explicitly by `bone_anchor`, a penalty force toward its frame-0 rest positions. The insertion is pinned by nothing: the tendon cap is given a *negative* stand-off so it overlaps the sclera, and points that share a B-spline stencil node share one grid velocity — a no-slip weld, for free. The honest caveat is that this weld is resolution-dependent and *cannot slip*, whereas a real tendon slides in its sheath. That is why the bellies are held about five cells clear of the globe; otherwise the whole arc of contact would weld and freeze the rotation.

8. The movie

Eight panels, black ground, lower-case parenthesised labels, one shared time cursor.

a)	anterior view	the cosmetic eye — white sclera, silvery iris, large black pupil, gold iridophore flecks, after the reference photograph — with the six muscles as their own material points, brightening with activation
b)	lateral view	the ovoid profile in the bony cup, the obliques, the trochlea
c)	strain	Green–Lagrange $\ E\ $ on a cut through globe <i>and</i> muscles
d)	stress	von Mises on the same cut
e)	grid momentum	$ v $ on the shared MLS-MPM grid — the medium through which a contracting muscle reaches the sclera
f)	activation	the six innervations against time
g)	length	each muscle as % of rest, with the globe’s own radius on the same axis: the muscles shorten, the eye must not
h)	gaze	horizontal / vertical / torsion against command, with the tracking error and its running mean, and the per-axis RMS printed

Panels c–e follow the codebase’s `grid.mp4` diagnostic, extended to two coupled bodies. Two rendering notes that are not cosmetic: the far hemisphere is culled, so the eye reads as a lit solid rather than a point cloud; and muscle points falling inside the globe’s exact orthographic silhouette *and* behind its centre are hidden, because a depth-sorted scatter alone lets a muscle show through the gaps between globe particles — most visibly straight across the black pupil.

9. What is deliberately not done

- **No promotion.** Eleven contracts and one implementation sit in `prototype/`. Plexus 2 requires evidence of reuse beyond the originating prototype first, and only `muscle_sleeve` and `mpm_scatter` [accumulate] look plausibly general.
- **`muscle_traction` and `muscle_insertion` are dead code, kept.** They are the *first* version — a line of action with a body force at the insertion. This note argues against them, and the argument should be checkable.
- **The integral term is implemented but off by default.** It is the oculomotor neural integrator and it is correct biology, but it was introduced in the same run as two other changes and contributed to a divergence. It has not been isolated since, and Phase 2 is where it should be.
- **No inverse pass on the anatomy.** Recovering the six insertion geometries from a gaze trace is the interesting Loop II question here, and it is untouched.